

Project

entitled

CYCLONE PREDICTION BASED ON SATELLITE DATA USING DEEP LEARNING

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Outlines

1. **Introduction**
2. **Objectives**
3. **Motivation**
4. **Literature Survey**
5. **Theoretical Background**
6. **Cyclone Prediction - Methodology and Workflow**
7. **Cyclone Prediction - Results**
8. **Path Forecasting – Methodology and Workflow**
9. **Path Forecasting - Results**
10. **Conclusion**
11. **Future Work**
12. **Our Contribution**
13. **References**
14. **Appendix - Dataset**

Introduction

- ❑ India, possess a large coastline of 7516.6 km is highly prone to cyclones.
- ❑ Cyclone prediction is a crucial aspect of meteorology and natural disaster management.
- ❑ Deep learning algorithms can be used to analyze and interpret a large amount of data from various sources, such as weather satellite imagery, atmospheric pressure readings, and wind patterns, to predict the likelihood and severity of a cyclone.

Objectives

- ❑ Our core objectives are:
 - ❑ Deep Learning model for detection of cyclone in its early stage of formation accurately
 - ❑ Predicting the direction and the path of the movement of the cyclones so as to alert the cyclone prone areas in advance.
 - ❑ To test the accuracy of latest object detection and time series forecasting methods.

Motivation

- ❑ Cyclones are natural disasters that can have devastating impacts on communities, causing loss of life, damage to property, and disruption to essential services.
- ❑ Around 10,000 people are killed on average each year due to cyclones. Gaja cyclone of 2018 killed 300 people and over 2500 buildings were destroyed.
- ❑ Accurate and reliable predictions of cyclone formation and development are therefore critical for helping people in affected areas to evacuate. Satellite data can provide valuable information about cyclones, such as their size, shape, and intensity.

Literature Review

- ❑ Md. Nazmul Haque *et al* [1] used two stage process to detect cyclone. In first step they detected the cyclone using various models like CNN , SVM, KNN,VGG16, RF, ANN and in second step they worked on detecting the eye of cyclone using YOLO algorithm.
- ❑ Preprocessing techniques like gray scaling the image and thresholding.
- ❑ Cross-validation techniques to prevent models from bias and overfitting.
- ❑ 87.86% mAP for cyclone detection

- ❑ Shakya S. *et al* [2] have used the deep learning techniques for two purposes, viz. classifying an image under the category of storm or no storm and predicting the storm location in the near future.
- ❑ 97% accuracy for storm no storm classification – CNN architecture.
- ❑ 84% confidence for detecting the cyclone – YOLO with R-CNN

Literature Review

- ❑ Selim Furkan Tekin et al. [3] studied tropical cyclone trajectories using deep learning techniques.
- ❑ RNN-based deep learning architecture called TrajGRU to capture complex temporal patterns and perform forecasts. The performance was analysed in terms of kilometers at various hourly resolutions.
- ❑ Used IBTrACS Dataset for predicting hurricane trajectories. This model also used atmospheric feature images at different pressure levels.

- ❑ Yanfei Zhang et al. [4] used matrix neural networks for cyclone track prediction for South Indian Ocean.
- ❑ Employed prominent RNN architectures as baseline models which includes Elman RNNs, Long Short-Term Memory and Gated Recurrent Unit Networks.
- ❑ Proposed use of MNNs to predict the future location of a cyclone based on its current position and various atmospheric variables such as temperature, humidity, and wind speed.

Theoretical Background

- ❑ Object Detection:
 - ❑ subfield of machine learning and computer vision that involves identifying and locating objects in images or videos
 - ❑ Successful object detection models : R-CNN and YOLO
 - ❑ Applications: autonomous vehicles, surveillance systems, and image/video search engines.
- R-CNN (Regional Based Convolutional Neural Network)
 - Uses CNN to classify and locate objects in an image
 - Generates a set of region proposals, which are potential locations of objects and outputs class probabilities and bounding box coordinates for each proposal
 - Computationally Expensive due to multiple forward and backward passes through the CNN for each image

YOLO

- ❑ YOLO (You Only Look Once) is a popular algorithm for object detection in images and videos.
- ❑ Developed by Joseph Redmon and Ali Farhadi in 2015.
- ❑ Fig.1 shows that YOLOv7 is much faster than other object detection models.

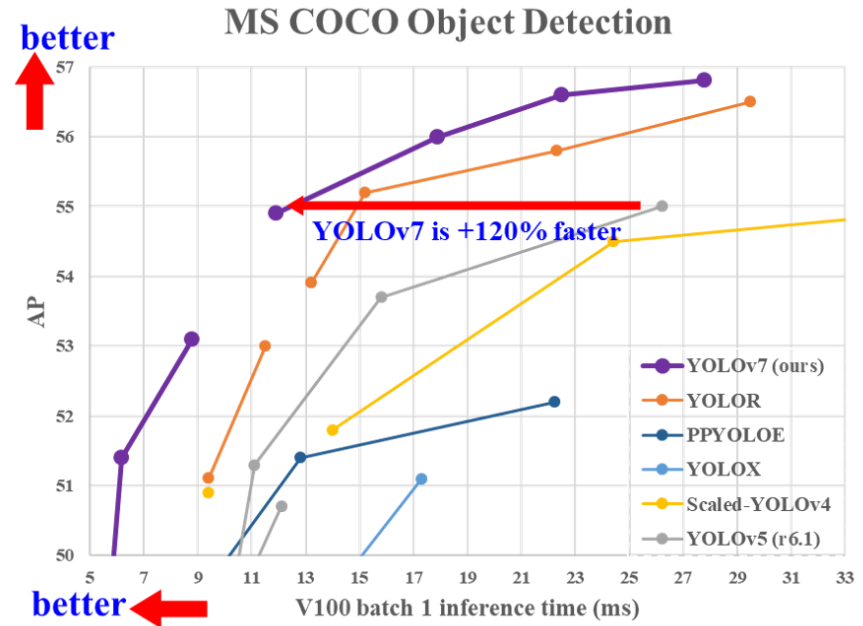


Fig:1 YOLO Speed Comparison [5]

YOLO Architecture

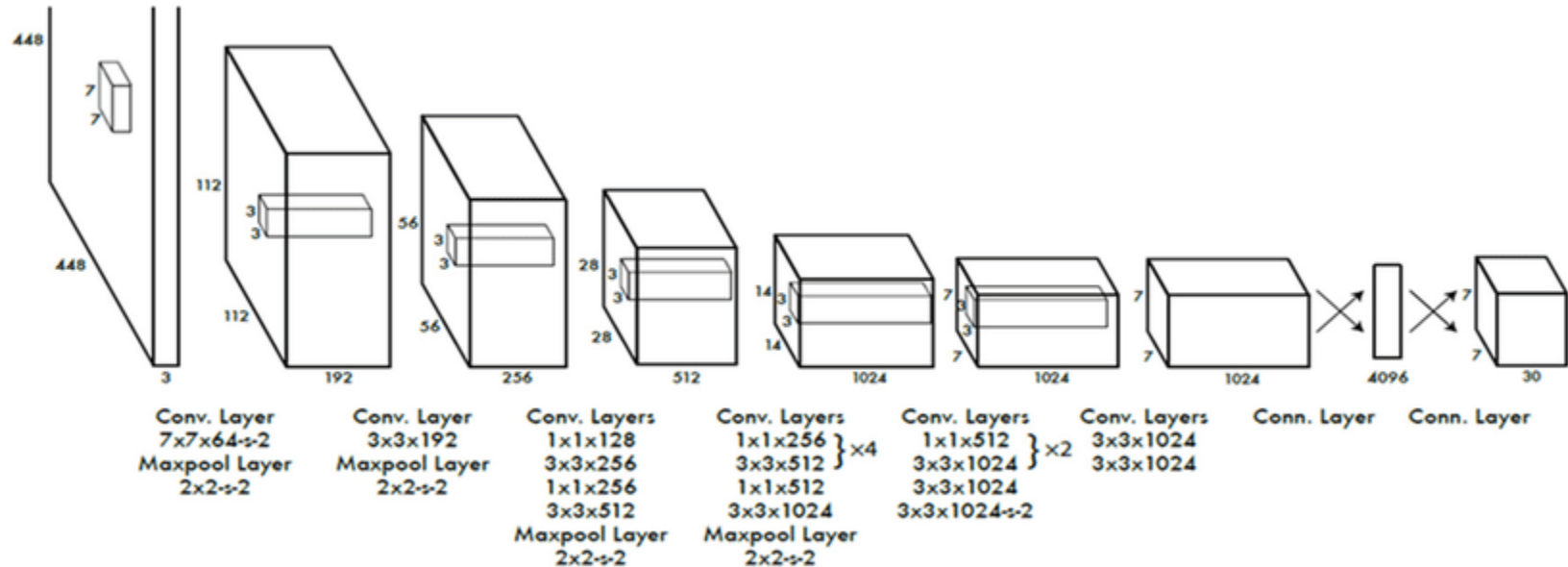


Fig:2 YOLO Architecture [5]

Tools and Libraries Used

- Google Colaboratory
- LabelImg
- Kaggle
- PyTorch
- OpenCV
- Matplotlib
- OS
- Keras
- GeoPandas

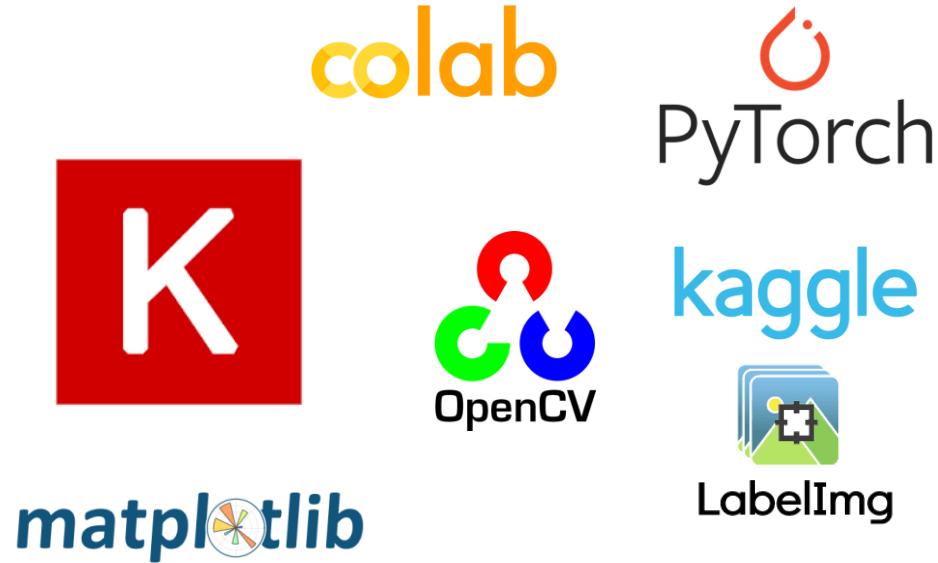


Fig:3 Toolkit

Workflow Diagram

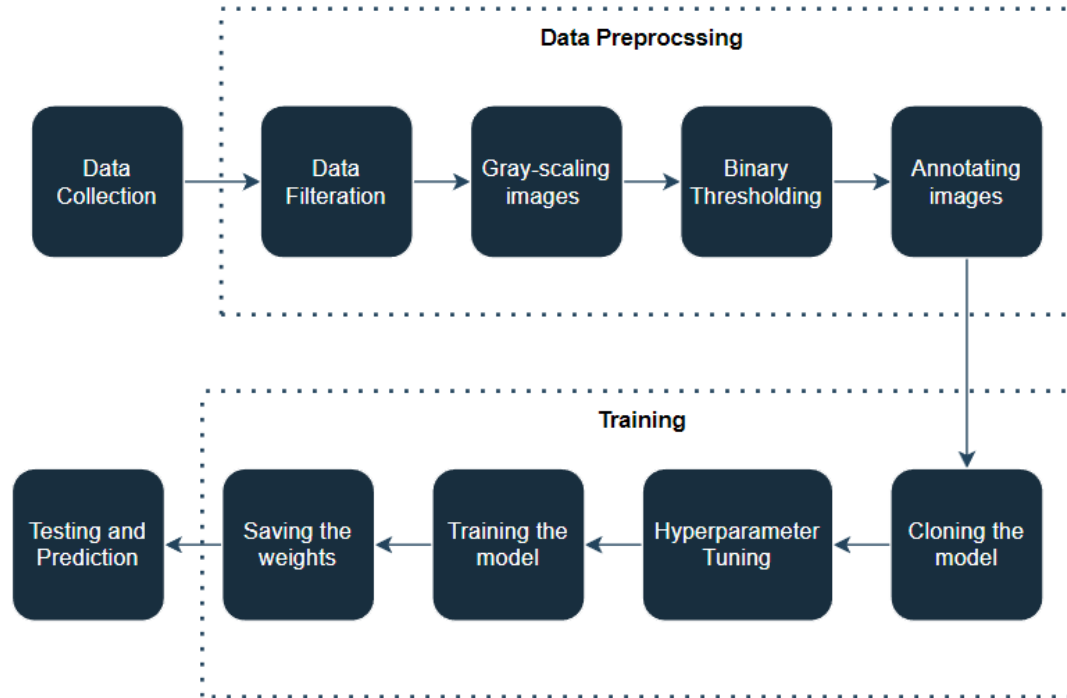
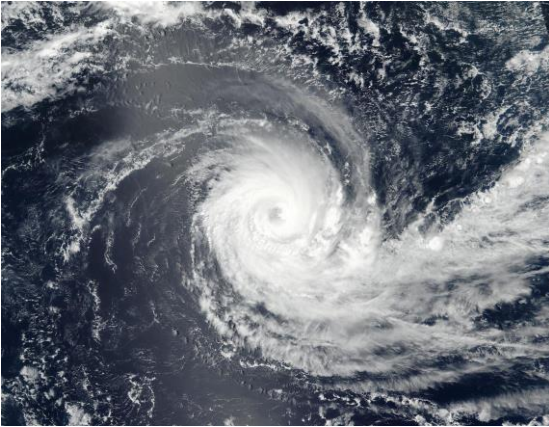
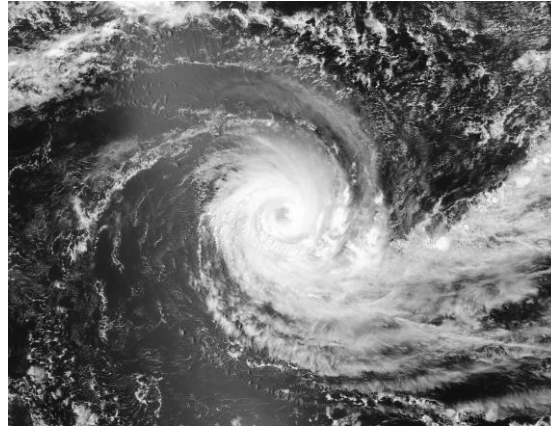


Fig:4 Workflow

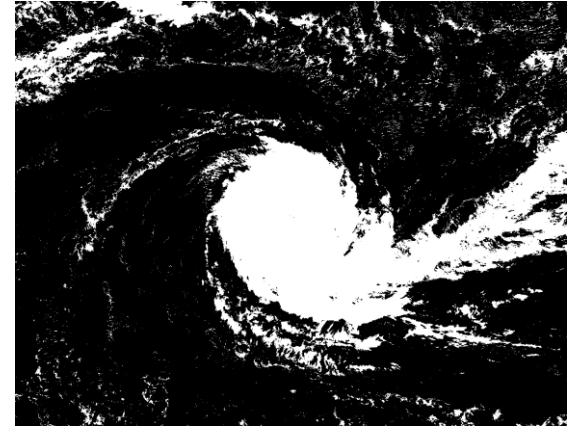
Data Preprocessing



a) Original Image



b) Grayscale Image



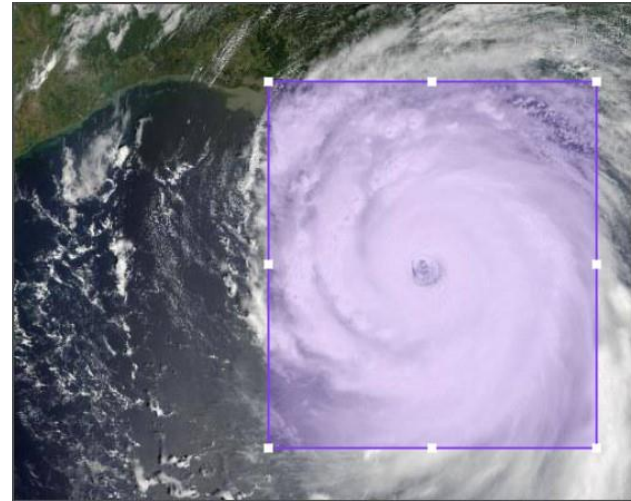
c) Binary Thresholding

Fig:5 Image Processing

Data Annotation



a) Original Image



b) Augmented Image

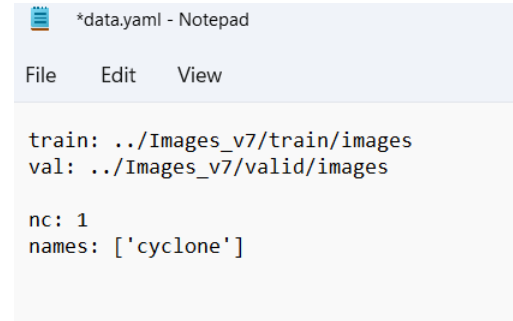
Fig:6 Annotations

Training the model

- **Setting up the model**
 - Download the original YOLOv7 model for training on our custom object
 - Modifying data.yaml file
- **Hyperparameter Tuning**

Sr. No.	Hyperparameter	Value
1	Batch Size	16
2	Epochs	500
3	Initial Learning Rate	0.05
4	Momentum	0.937
5	Weight Decay	0.0005

Table:1 Hyperparameters



```
*data.yaml - Notepad
File Edit View

train: ../Images_v7/train/images
val: ../Images_v7/valid/images

nc: 1
names: ['cyclone']
```

Fig:7 data.yaml file

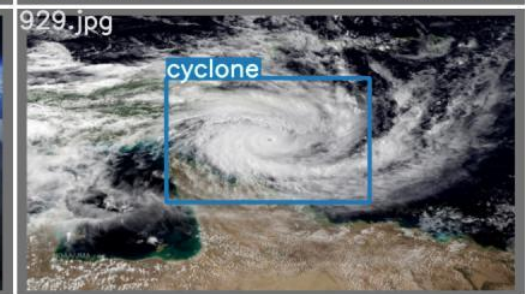


Fig:8 Detected Cyclone Images

Training Results

	True Positive	True Negative
Predicted Positive	320	27
Predicted Negative	33	22

Table:2 Confusion Matrix

Sr. No.	Metrics	Training	Testing
1	class	cyclone	cyclone
2	Images	1608	402
3	mAP	0.742	0.68
4	Accuracy	0.91	0.85

Table:3 Accuracy

Cyclone Path Prediction

- ❑ Time Series Forecasting:
 - ❑ Using statistical models and machine learning algorithms to make predictions about future values based on historical data.
 - ❑ A time series is a sequence of data points collected at regular intervals over time.
 - ❑ Examples of time series data include stock prices, weather data, and sensor data from industrial equipment or latitudes, longitudes, and other features of a cyclone's eye at regular intervals, which is being used in this project.
 - ❑ Techniques include statistical methods such as ARIMA as well as machine learning algorithms such as neural networks.

Long Short-Term Memory

- A traditional RNN has a single hidden state that is passed through time, which can make it difficult for the network to learn long-term dependencies.
- LSTMs address this problem by introducing a memory cell, which is a container that can hold information for an extended period of time.
- The memory cell is controlled by three gates: the input gate, the forget gate, and the output gate.
- These gates decide what information to add to, remove from, and output from the memory cell.

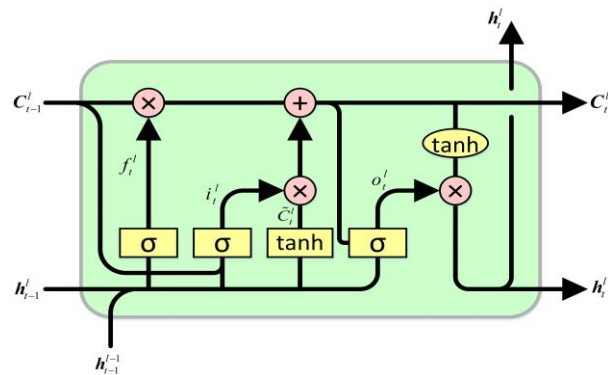


Fig:10 LSTM architecture [9]

Gated-Recurrent Unit

- The basic idea behind GRU is to use gating mechanisms to selectively update the hidden state of the network at each time step.
- The gating mechanisms are used to control the flow of information in and out of the network.
- The GRU has two gating mechanisms, called the reset gate and the update gate.
- The reset gate determines how much of the previous hidden state should be forgotten, while the update gate determines how much of the new input should be used to update the hidden state.

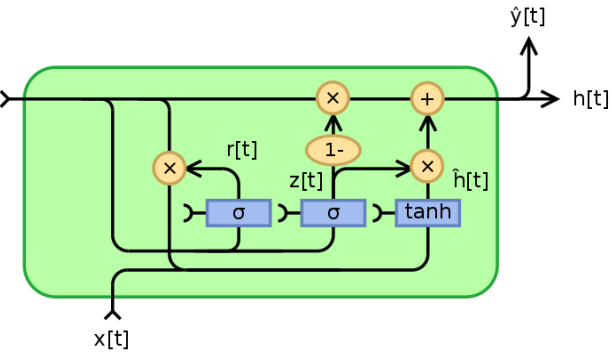
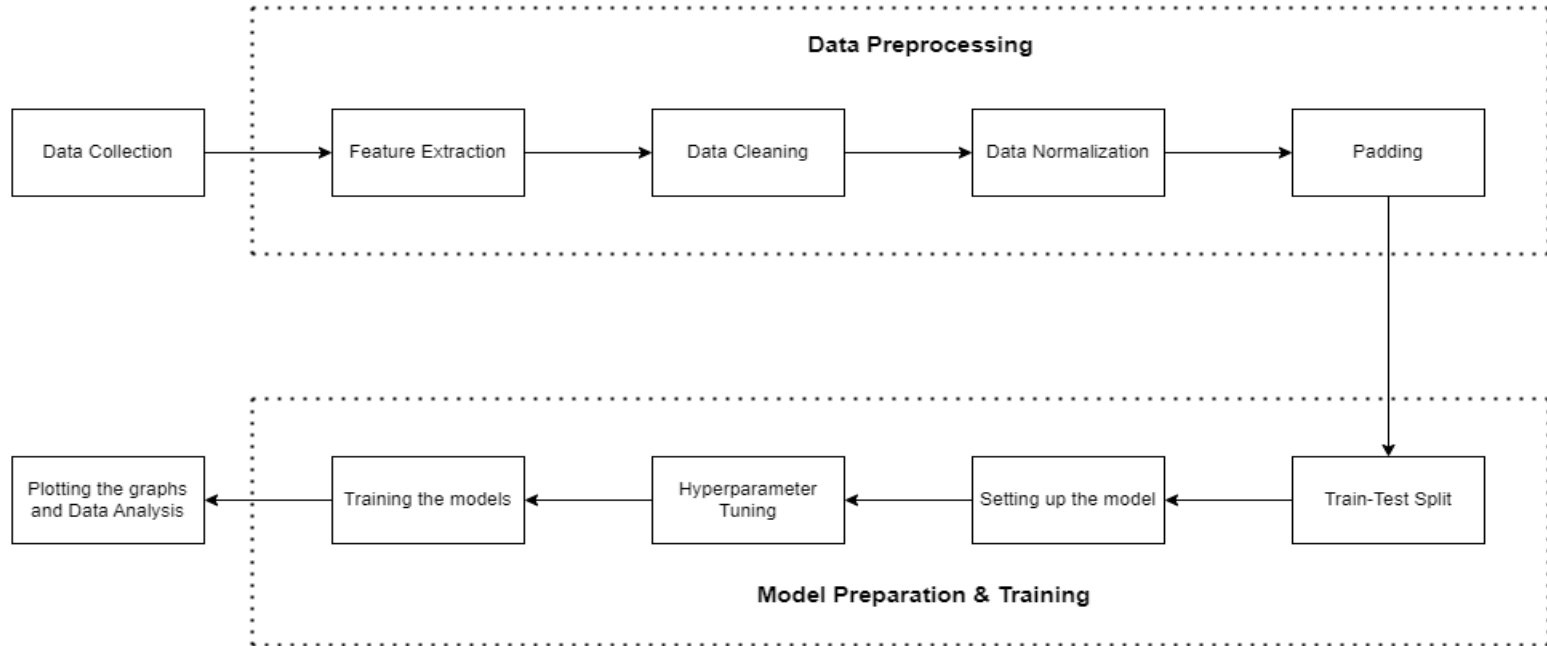


Fig:11 GRU architecture [10]

Workflow Diagram



Dataset - IBTrACS

SID	SEASON	NUMBER	BASIN	SUBBASIN	NAME	ISO_TIME	NATURE	LAT	LO_N	...	BOM_GUST_PER	REUNION_GUST	REUNION_GUST_PER	USA_SEAHGT	USA_SEARAD_NE	USA_SEARAD_SE	USA_SEARAD_SW	USA_SEARAD_NW	STORM_SPEED	STORM_DTR
Year								degrees_north	degrees_east	...	second	kts	second	ft	nmile	nmile	nmile	nmile	kts	degrees
1842298N11080	1842	1	NI	BB	NOT_NAMED	1842-10-25 03:00:00	NR	10.9	80.3	...									9	286
1842298N11080	1842	1	NI	BB	NOT_NAMED	1842-10-25 06:00:00	NR	10.8709	79.8265	...									9	287
1842298N11080	1842	1	NI	BB	NOT_NAMED	1842-10-25 09:00:00	NR	10.8431	79.3524	...									9	287
1842298N11080	1842	1	NI	BB	NOT_NAMED	1842-10-25 12:00:00	NR	10.8188	78.8772	...									9	287
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2022349N13068	2022	87	NI	AS	NOT_NAMED	16-12-2022 18:00	NR	14.2	62.9	...									7	259
2022349N13068	2022	87	NI	AS	NOT_NAMED	16-12-2022 21:00	NR	14.115	62.5445	...									7	254
2022349N13068	2022	87	NI	AS	NOT_NAMED	17-12-2022 00:00	NR	14.0	62.2	...									7	252
2022349N13068	2022	87	NI	AS	NOT_NAMED	17-12-2022 03:00	NR	13.8945	61.8416	...									7	254
2022349N13068	2022	87	NI	AS	NOT_NAMED	17-12-2022 06:00	NR	13.7983	61.4728	...									7	255

rows x 183 columns

Data Preprocessing

- Feature Extraction
 - Dataset used for cyclone path prediction initially has lot of features that are of no use to us like time, year and nature of cyclone.
 - We only extracted ID, latitude , longitude, wind speed and wind direction only
- Data Cleaning
 - Few values of latitude, longitude were in string format which was converted into float format
 - The rows with missing or null values were removed from the dataset

Data Preprocessing

- Data Normalization
 - In order to achieve better accuracy of the model we divide the latitude, longitude, wind speed and wind direction by their maximum values.
- Data Padding
 - In order to make sure all cyclones have same number of time stamps we pad the data by taking mean of data
 - Cyclones having number of time stamps less than 18 and greater than 35 are ignored.

Data

```
[array(['1845336N10074', 0.1111, 0.8256, 0.0625, 0.7861], dtype=object),  
array(['1845336N10074', 0.1115, 0.8239, 0.0625, 0.7889], dtype=object),  
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array(['1845336N10074', 0.1439, 0.7949, 0.0833, 0.9806], dtype=object),  
array(['1845336N10074', 0.1459, 0.7947, 0.0833, 0.9833], dtype=object),  
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```

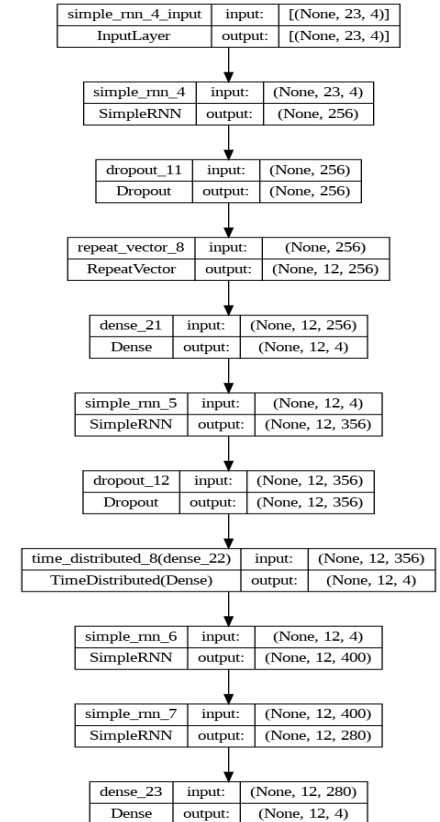
Training Model

- For training data, 75% of the original dataset is used and the other 25% is used for test split.
- For every cyclone 23 datapoints are used as input and 12 datapoints are predicted by the forecasting model.
- Datapoints are recorded at the intervals of 3 hours.
- This implies almost 3 days of data is used to predict the path of cyclone for next two days

Model Description – Simple RNN

Sr. No.	Hyperparameters	Value
1	Batch Size	32
2	Epochs	500
3	Optimizer	Adam
4	Loss Function	Mean Square Error
5	Activation Function	Relu

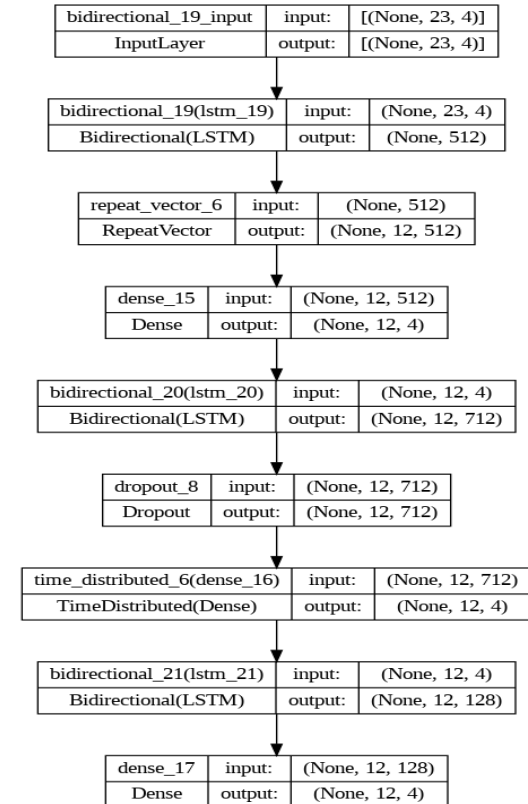
Table:4 Hyperparameters of Simple RNN



Model Description – Bi-LSTM

Sr. No.	Hyperparameters	Value
1	Batch Size	32
2	Epochs	500
3	Optimizer	Adam
4	Loss Function	Mean Square Error
5	Activation Function	Tanh

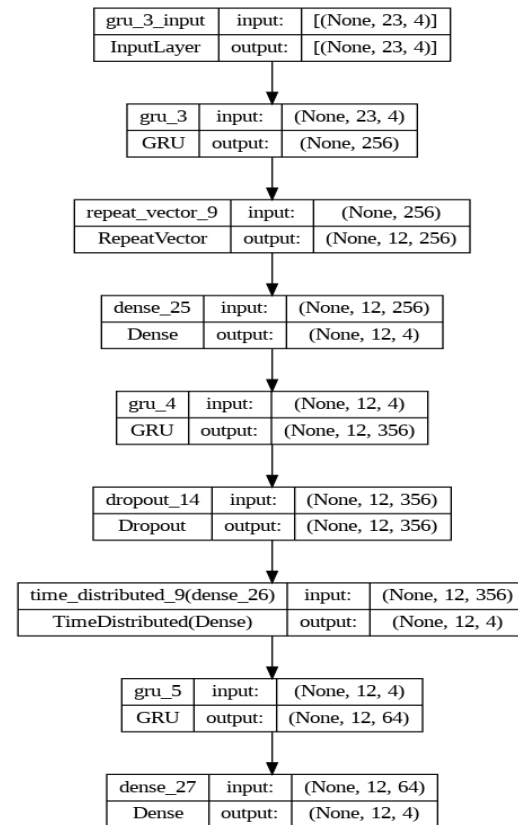
Table:5 Hyperparameters of Bi-LSTM



Model Description – GRU

Sr. No.	Hyperparameters	Value
1	Batch Size	32
2	Epochs	500
3	Optimizer	Adam
4	Loss Function	Mean Square Error
5	Activation Function	Tanh

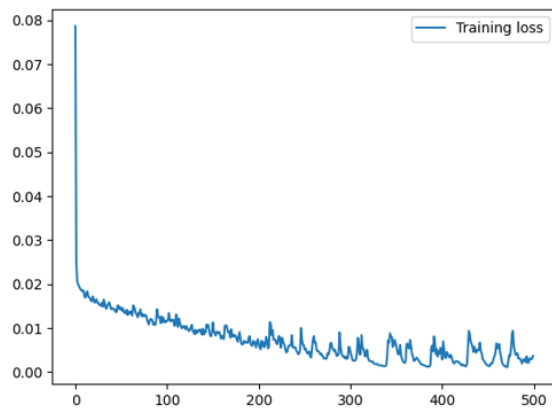
Table:6 Hyperparameters of GRU



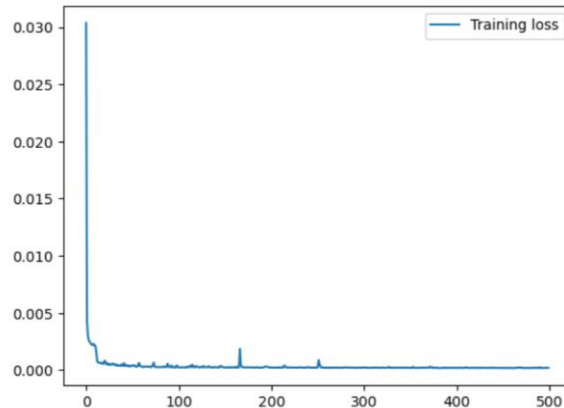
Challenges Faced

- **Threshold Parameter Tuning**
- **Uneven Length of Cyclone Data**
- **Batch size and Memory Limit**
- **Hyperparameter Tuning**
- **Problem of Overfitting**
- **Limited support of GPU**

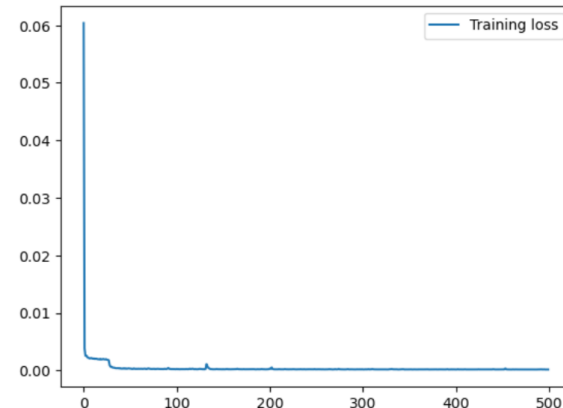
Training Loss vs Epochs



(a)



(b)



(c)

Fig 18: Training Loss vs Epochs

RNN Model

Training Loss: 5.07×10^{-4}

Bi-LSTM Model

Training Loss: 1.842×10^{-4}

GRU Model

Training Loss: 1.579×10^{-4}

Mean Squared Error Loss

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Equation 1: MSE

Model	6 hour	12 hour	18 hour	24 hour
Simple RNN	0.4374	0.5516	0.687	1.5367
Bi-LSTM	0.4799	0.5322	0.654	1.3789
GRU	0.4932	0.5055	0.632	1.1703

Table 7: Mean Squared Error

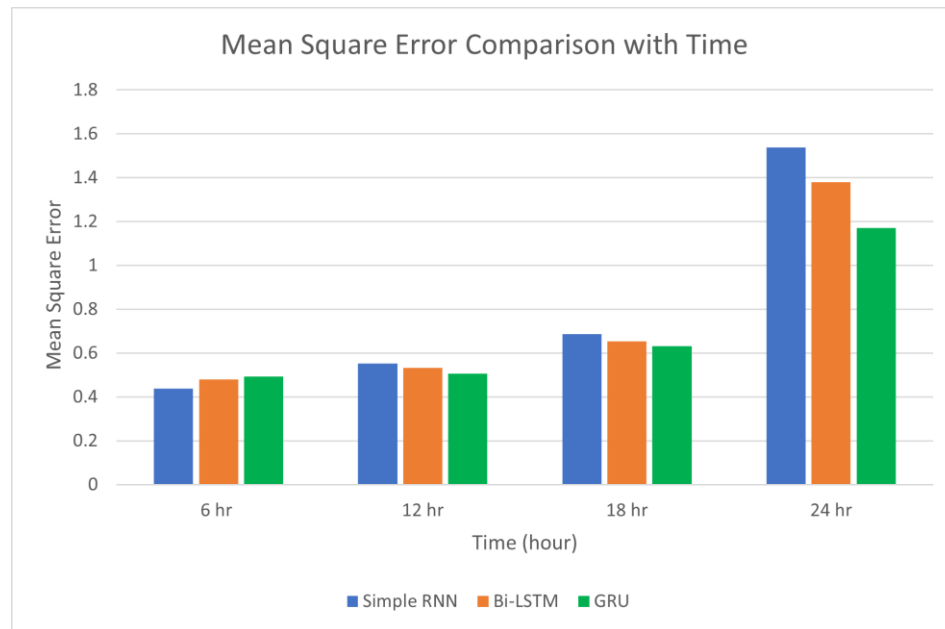


Fig 19: Mean Squared Error – Graphical Representation

Distance-wise km Loss

- Haversine Formula to calculate distance between two co-ordinates (in kilometre)

$$2r \arcsin \left(\sqrt{\sin^2 \left(\frac{\varphi_2 - \varphi_1}{2} \right) + \cos \varphi_1 \cdot \cos \varphi_2 \cdot \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right).$$

Equation 2: Haversine Formula

where

φ_1 and φ_2 are latitudes

λ_1 and λ_2 are the longitudes

- Library used- `geopy.distance`
- Mean of these values are calculated for future 6hr, 12hr, 18hr, 24hr and 36hr time

Distance-wise Model Comparison

Model	6 hour (km)	12 hour(km)	18 hour(km)	24 hour(km)	36 hour(km)
Simple RNN	40.87	104.56	181.33	190.56	202.32
Bi-LSTM	138.54	143.33	165.44	170.39	212.43
GRU	119.90	127.05	162.27	165.32	197.099

Table 8: Distance wise model comparison error

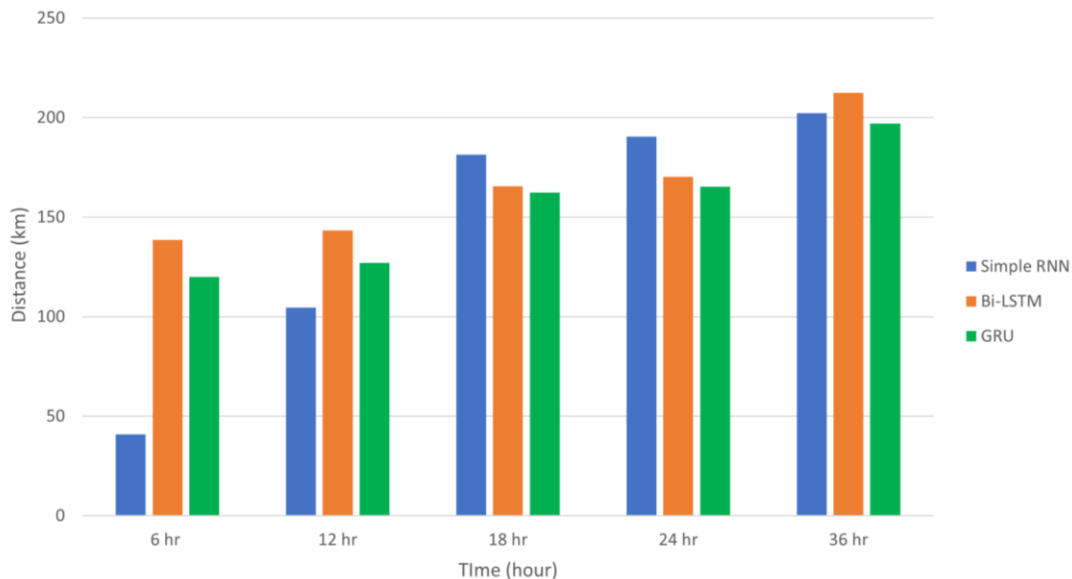


Fig 20: Distance wise errors- Graphical Representation

Model Comparison Graphs

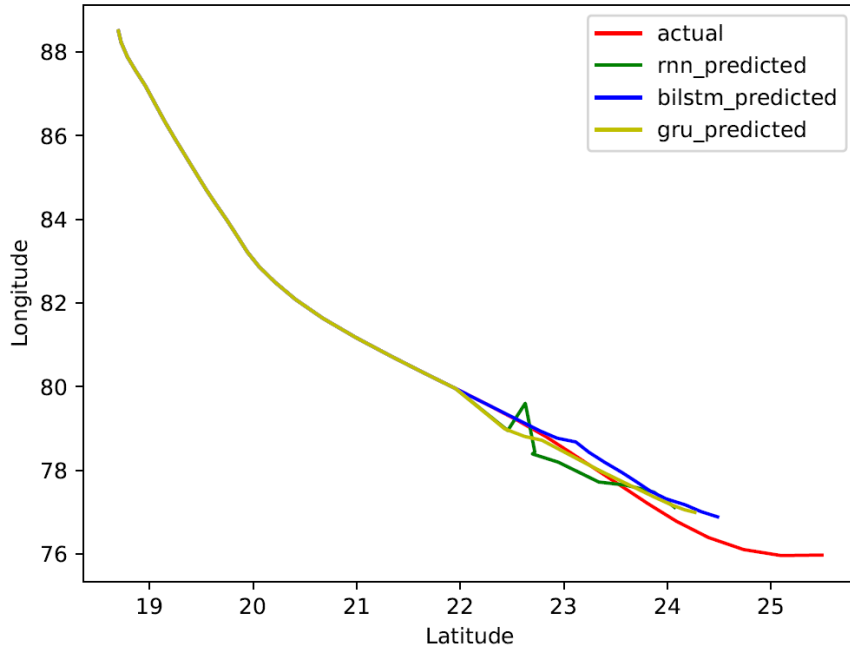


Fig 21: Test result 1

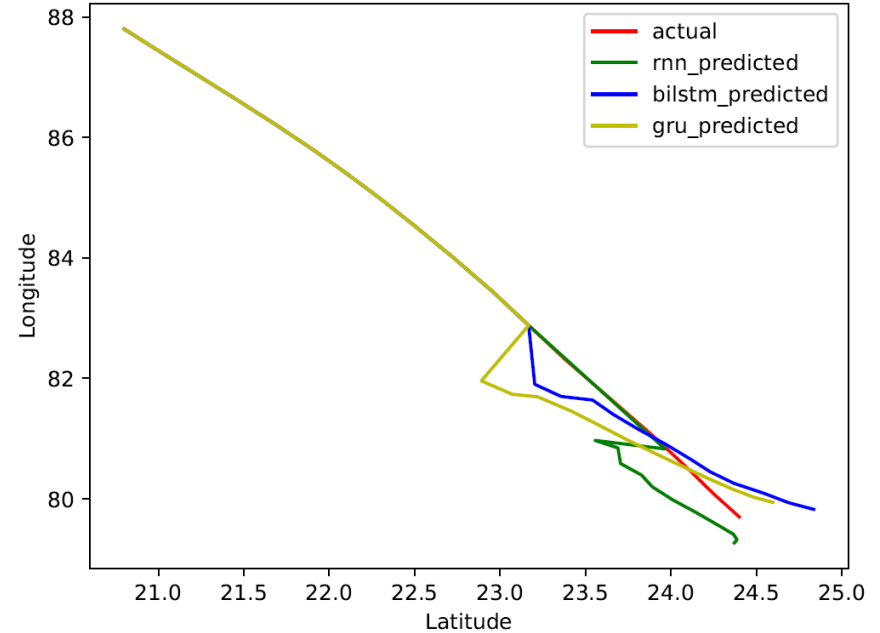


Fig 22: Test result 2

Model Comparison

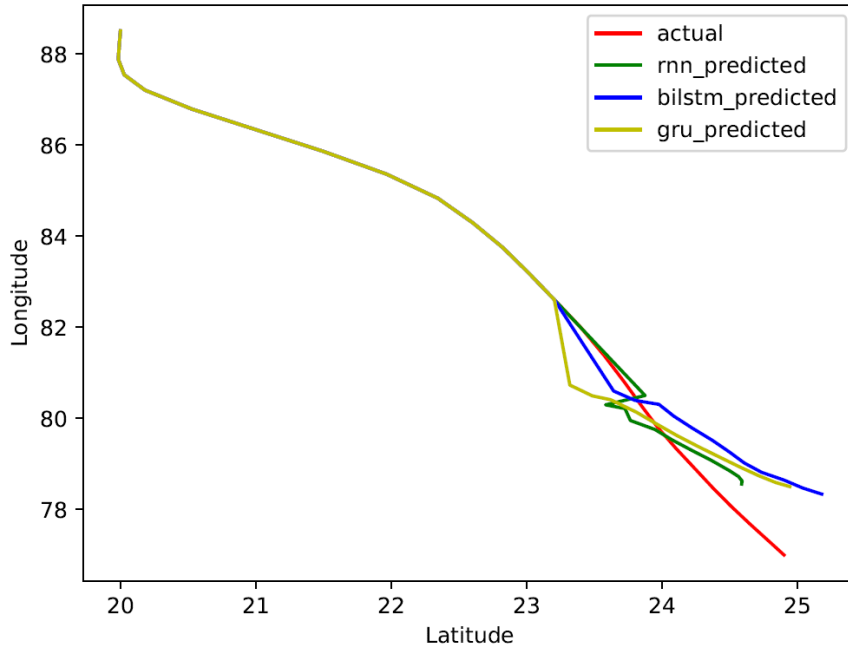


Fig 23: Test result 3

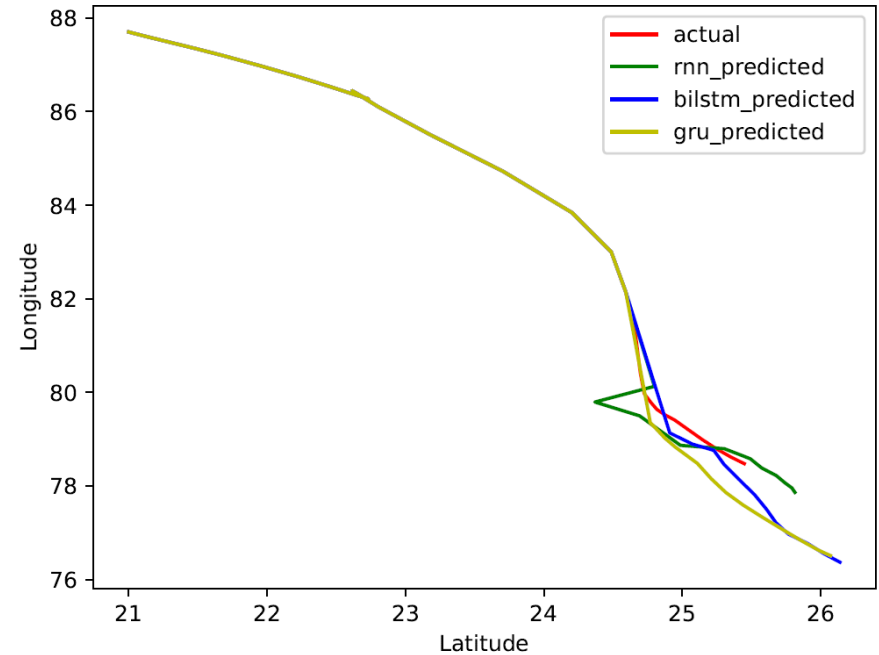


Fig 24: Test result 4

Model Comparison

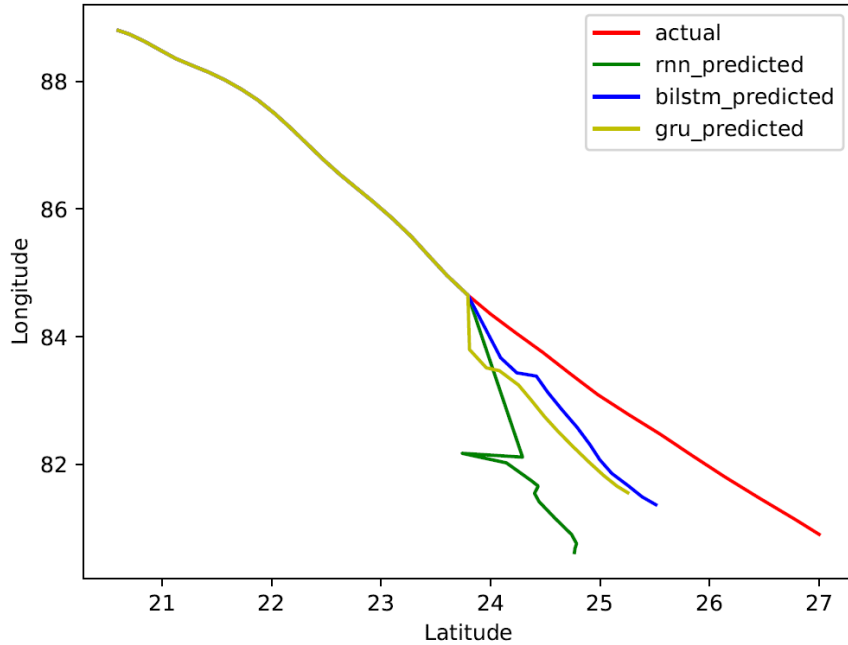


Fig 25: Test result 5

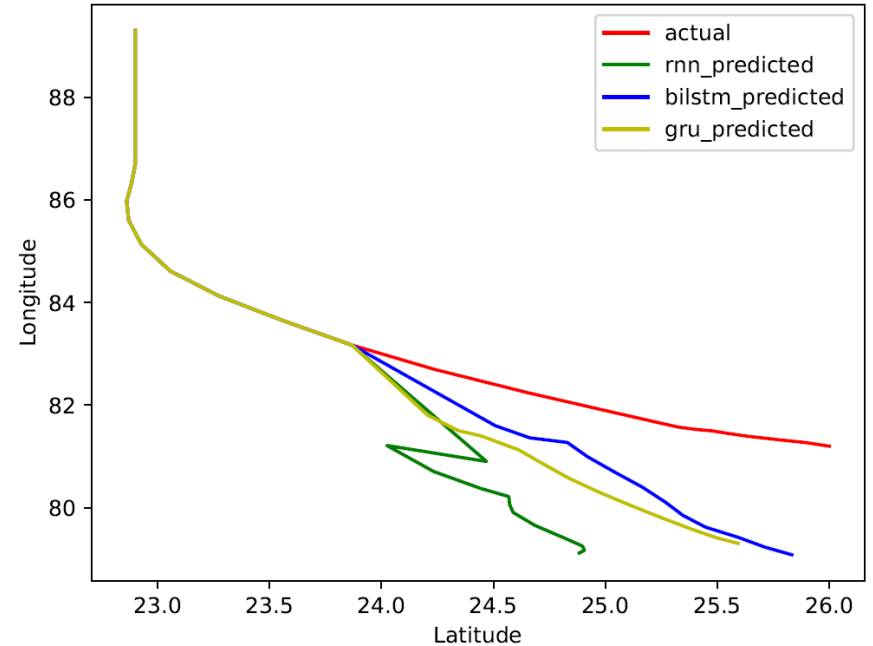


Fig 26: Test result 6

RNN Model – India Plot

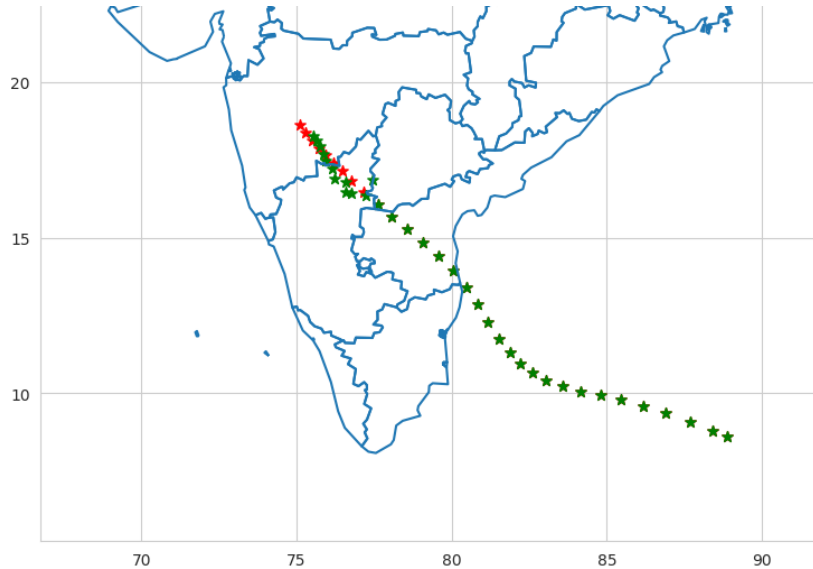


Fig 27: RNN Test result I on Map

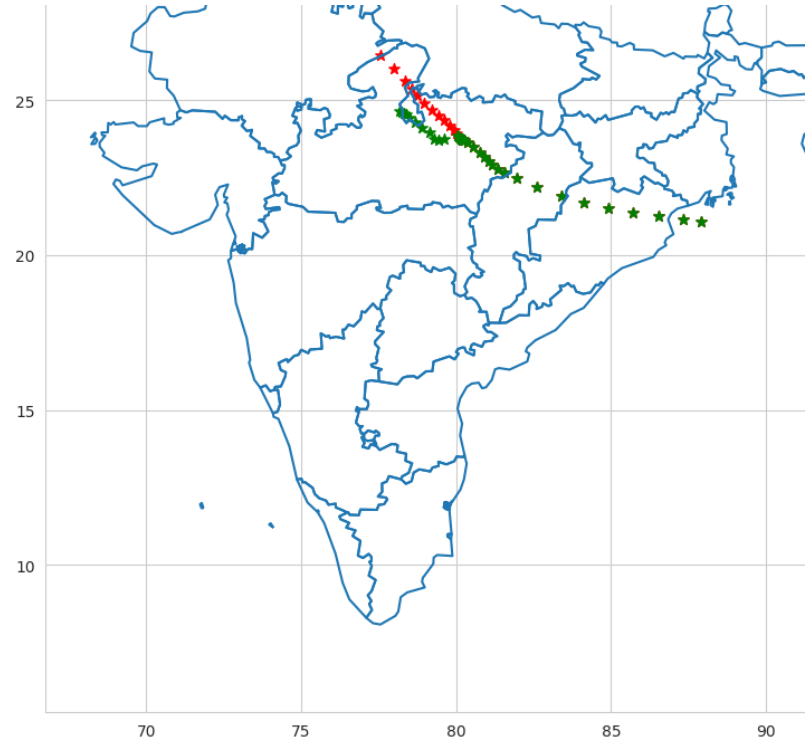


Fig 28: RNN Test result II on Map

Bi-LSTM Model – India Plot

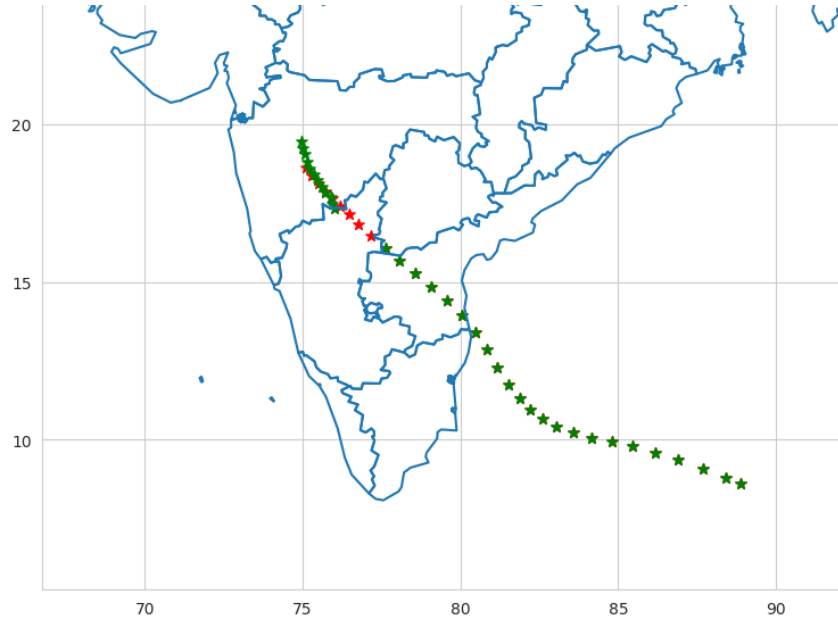


Fig 29: Bi-LSTM Test result I on Map

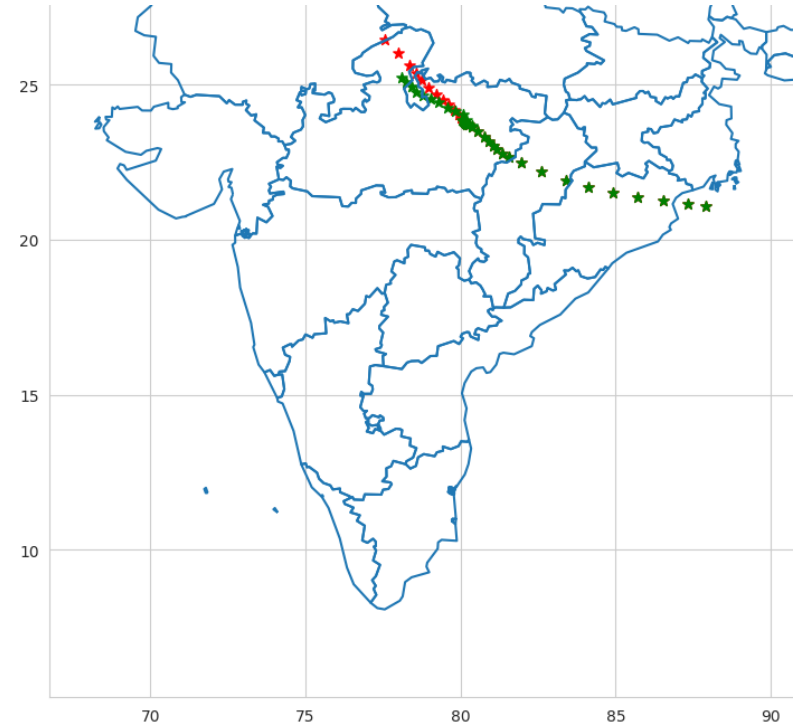


Fig 30: Bi-LSTM Test result II on Map

GRU Model – India Plot

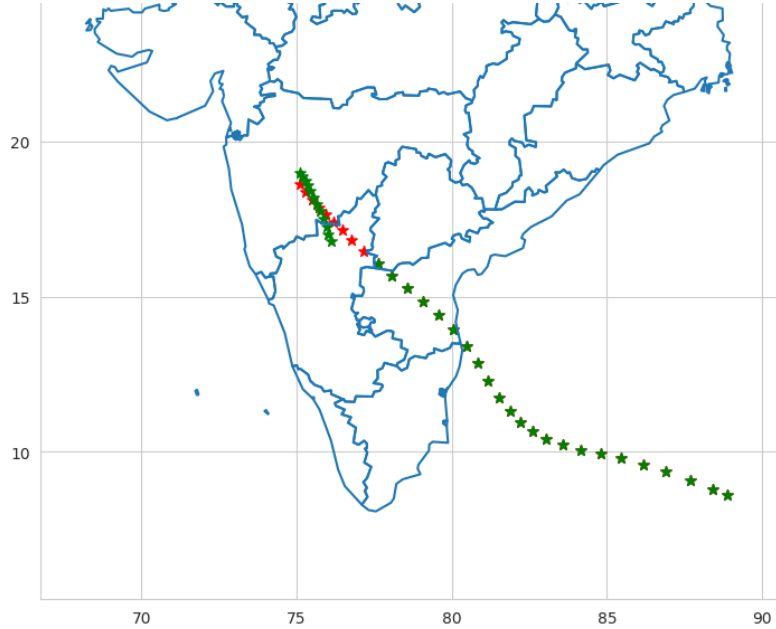


Fig 31: GRU Test result I on Map

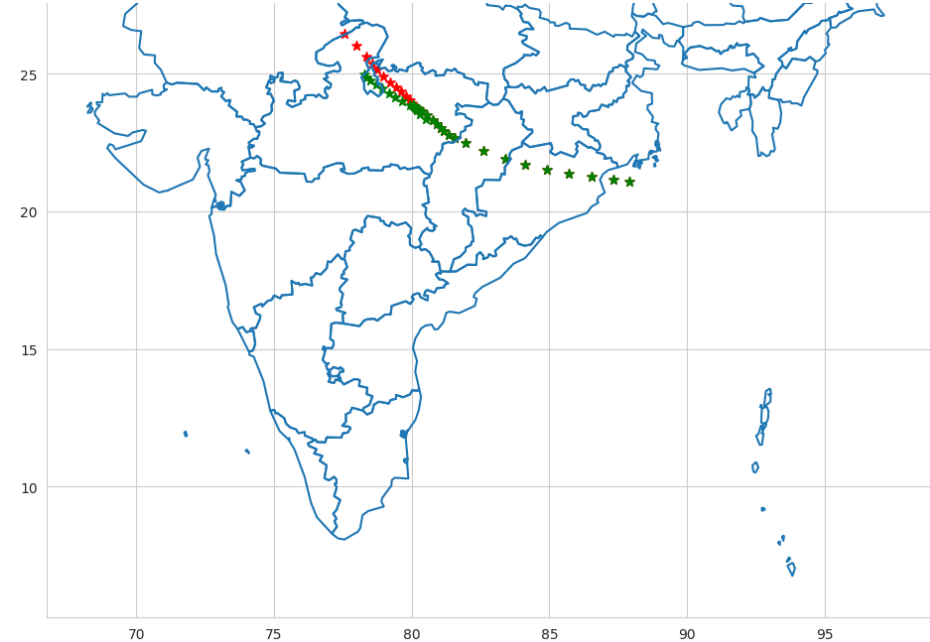


Fig 32: GRU Test result II on Map

Conclusion

- YOLOv7 model for cyclone detection was discussed which was trained on 1608 and tested on 402 annotated images.
- This model produced mAP of 0.742 and 0.68 on training and testing data with accuracy of 0.91 (training) and 0.85 (testing)
- 3 Deep Neural Network models – RNN, Bi-LSTM and GRU networks were used for path forecasting model
- Mean square error, distance mean square error were calculated and plotted on India map for better interpretation of the regions affected

“RNN model was effective for short term (3hr & hr) prediction whereas GRU network model was having least error and showing better results for long term predictions (12-36 hr)”

Future Work

- **Refining the cyclone detection model by increasing the dataset and the confidence level**
- **Use of more advanced and latest frameworks of YOLO and their comparison for better accuracy and results**
- **Architectural changes in the cyclone path prediction models and use of some more deep learning models for better and accurate results with least error**

Our Contribution

- We have used YOLOv7 for Cyclone Detection Model instead of YOLOv3
- We have used 4 features, Latitudes, Longitudes as well as Wind Speed and Wind direction to train our model
- Comparison of three models RNN, Bi-LSTM and GRU with hyperparameter tuning and Distance wise results obtained.

References

- [1] **Journal** M. Haque, A. Adel, and K. Alam, “Deep learning techniques in cyclone detection with cyclone eye localization based on satellite images,” 12 2021.
- [2] **Journal** S. Shakya, S. Kumar, and M. Goswami, “Deep learning algorithm for satellite imaging based cyclone detection,” IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 13, pp. 827–839, 2020. Journal of Remote Sensing. Vol. 6, No. 1, 2018, pp. 1-5. doi: 10.11648/j.ajrs.20180601.11]
- [3] F. Selim Furkan Tekin, “Forecasting of tropical cyclone trajectories with deep learning.” [Online]. Available: <https://sftekin.github.io/pdfs/hurricane.pdf>
- [4] Y. Zhang, R. Chandra, and J. Gao, “Cyclone track prediction with matrix neural networks,” in 2018 International Joint Conference on Neural Networks (IJCNN), 2018, pp. 1–8
- [5] <https://github.com/WongKinYiu/yolov7>
- [6] <https://learnopencv.com/yolov7-object-detection-paper-explanation-and-inference/>
- [7] <https://www.kaggle.com/datasets/mikolajbabula/disaster-images-dataset-cnn-model>
- [8] “Recurrent neural networks cheatsheet.” [Online]. Available: <https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks>
- [9] https://en.wikipedia.org/wiki/Long_short-term_memory
- [10] “Gated recurrent unit.” [Online]. Available: https://en.wikipedia.org/wiki/Gated_recurrent_unit
- [11] S. Alemany, J. Beltran, A. Perez, and S. Ganzfried, “Predicting hurricane trajectories using a recurrent neural network,” 2018.

Appendix 1 - Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	SID	SEASON	NUMBER	BASIN	SUBBASIN	NAME	ISO_TIME	NATURE	LAT	LON	WMO_WII	WMO_PRE	WMO_AGI	TRACK_TY	DIST2LANI	LANDFALL	IFLAG	USA_AGEN	USA_ATCF	USA_LAT	USA_LON	USA_RECC	USA_STAT	USA
2		Year							degrees_n	degrees_e	kts	mb			km	km				degrees_n	degrees_e			kts
3	1842298N	1842	1	NI	BB	NOT_NAM	1842-10-2	NR	10.9	80.3				main	43	0								
4	1842298N	1842	1	NI	BB	NOT_NAM	1842-10-2	NR	10.8709	79.8265				main	0	0								
5	1842298N	1842	1	NI	BB	NOT_NAM	1842-10-2	NR	10.8431	79.3524				main	0	0								
6	1842298N	1842	1	NI	BB	NOT_NAM	1842-10-2	NR	10.8188	78.8772				main	0	0								
7	1842298N	1842	1	NI	BB	NOT_NAM	1842-10-2	NR	10.8	78.4				main	0	0								
8	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.7884	77.9194				main	0	0								
9	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.7845	77.4299				main	0	0								
10	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.7884	76.9256				main	0	0								
11	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.8	76.4				main	0	0								
12	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.8096	75.8424				main	10	10								
13	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.8274	75.2649				main	65	65								
14	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.8565	74.6799				main	119	119								
15	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.9	74.1				main	171	171								
16	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	10.9645	73.5308				main	216	216								
17	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.0423	72.9702				main	264	264								
18	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.1239	72.4245				main	318	318								
19	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.2	71.9				main	362	362								
20	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.2527	71.4073				main	406	406								
21	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.2994	70.9348				main	456	456								
22	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.3464	70.4698				main	497	497								
23	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.4	70				main	545	545								
24	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.4762	69.4627				main	594	594								
25	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.5579	68.9253				main	655	655								
26	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.6356	68.4252				main	708	708								
27	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.7	68				main	743	743								
28	1842298N	1842	1	NI	AS	NOT_NAM	1842-10-2	NR	11.7348	67.6058				main	774	774								

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Appendix 1 - Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
60433	2022342N	2022	85 NI	BB	MANDOU(#####	NR			10.9	81.4				PROVISIO	163	152 O_____	tcvitals	BB062022	10.9	81.4			
60434	2022342N	2022	85 NI	BB	MANDOU(#####	NR			11.0274	81.3549				PROVISIO	163	163 P_____		BB062022	11.0274	81.3549			
60435	2022342N	2022	85 NI	BB	MANDOU(#####	NR			11.2	81.4				PROVISIO	163	156 O_____	tcvitals	BB062022	11.2	81.4			
60436	2022342N	2022	85 NI	BB	MANDOU(#####	NR			11.5125	81.3374				PROVISIO	156	127 P_____		BB062022	11.5125	81.3374			
60437	2022342N	2022	85 NI	BB	MANDOU(#####	NR			11.9	81.2				PROVISIO	127	89 O_____	tcvitals	BB062022	11.9	81.2			
60438	2022342N	2022	85 NI	BB	MANDOU(#####	NR			12.3001	80.9752				PROVISIO	89	54 P_____		BB062022	12.3001	80.9752			
60439	2022342N	2022	85 NI	BB	MANDOU(#####	NR			12.7	80.7				PROVISIO	54	O_____	tcvitals	BB062022	12.7	80.7			
60440	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.5	67.8				PROVISIO	677	677 O_____	tcvitals	AS072022	13.5	67.8			
60441	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.7126	67.5076				PROVISIO	697	697 P_____		AS072022	13.7126	67.5076			
60442	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.9	67.2				PROVISIO	716	716 O_____	tcvitals	AS072022	13.9	67.2			
60443	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14.0374	66.865				PROVISIO	741	741 P_____		AS072022	14.0374	66.865			
60444	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14.1	66.5				PROVISIO	777	777 O_____	tcvitals	AS072022	14.1	66.5			
60445	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14.0725	66.1074				PROVISIO	817	817 P_____		AS072022	14.0725	66.1074			
60446	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14	65.7				PROVISIO	862	862 O_____	tcvitals	AS072022	14	65.7			
60447	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.9651	65.2701				PROVISIO	902	902 P_____		AS072022	13.9651	65.2701			
60448	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.9	64.9				PROVISIO	947	938 O_____	tcvitals	AS072022	13.9	64.9			
60449	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.6973	64.6998				PROVISIO	945	936 P_____		AS072022	13.6973	64.6998			
60450	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.6	64.5				PROVISIO	936	889 O_____	tcvitals	AS072022	13.6	64.5			
60451	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.8776	64.1305				PROVISIO	881	826 P_____		AS072022	13.8776	64.1305			
60452	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14.2	63.7				PROVISIO	826	787 O_____	tcvitals	AS072022	14.2	63.7			
60453	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14.26	63.2923				PROVISIO	787	764 P_____		AS072022	14.26	63.2923			
60454	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14.2	62.9				PROVISIO	764	749 O_____	tcvitals	AS072022	14.2	62.9			
60455	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14.115	62.5445				PROVISIO	742	730 P_____		AS072022	14.115	62.5445			
60456	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			14	62.2				PROVISIO	730	716 O_____	tcvitals	AS072022	14	62.2			
60457	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.8945	61.8416				PROVISIO	712	703 P_____		AS072022	13.8945	61.8416			
60458	2022349N	2022	87 NI	AS	NOT_NAM #####	NR			13.7983	61.4728				PROVISIO	703	O_____	tcvitals	AS072022	13.7983	61.4728			
60459																							
60460																							

THANK YOU !